WASP-10b

R-0696

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-10_231110` · created 2026-07-27T21:14:43+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460258.7929 +/- 0.006 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.065 +/- 0.06
Transit depth (Rp/Rs)^2	0.43 +/- 0.79 [%]
Orbital Inclination (inc)	84.86 +/- 1.87
Airmass coefficient 1 (a1)	0.98 +/- 0.65
Airmass coefficient 2 (a2)	0.05 +/- 0.23
Scatter in the residuals of the lightcurve fit is	70.09 %
Best Comparison Star	#2 - [466, 438]
Optimal Aperture	2.77
Optimal Annulus	3.75
Transit Duration (day)	0.032 +/- 0.032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.065 ± 0.06 vs 0.1592 (deviation 1.56σ)
Duration fit vs published	0.032 d vs 0.0946 d (deviation 1.94σ)
Mid-time O–C	14.3 min
Depth SNR	1.1
Residual scatter	70.09 %

Light curve

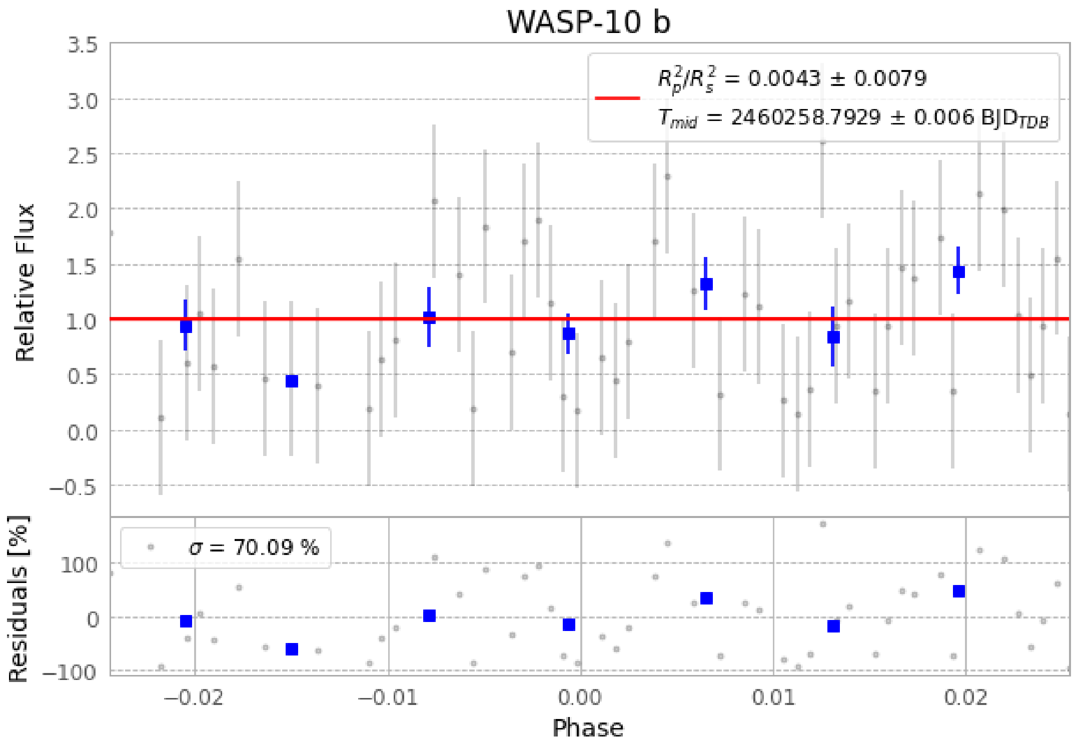

Figure 1 — FinalLightCurve_WASP-10 b_2023-11-09.png (SHA-256 0a0a6f1019791229…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [152, 246], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 ddd54faa04fdd9cc…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813`

Data provenance

76 FITS frames (50 MB), `WASP-10231110050615.FITS` → `WASP-10231110085115.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-231110.log (0c56eb4996b7…), FinalLightCurve_WASP-10 b_2023-11-09.png (0a0a6f101979…), FinalLightCurve_WASP-10 b_2023-11-09.csv (835bb24c06e7…)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-27 21:14Z.